Department of Mathematics, IIT Madras

MA2031 Linear Algebra July - November 2023

**Assignment 3**

1. Suppose the entries in each row of an $n \times n$ matrix $A$ add up to $s$. Find an eigenvector of $A$.

2. Suppose the entries in each column of an $n \times n$ matrix $A$ add up to $s$. Describe how to obtain an eigenvector of $A$.

3. Find all eigenvalues and corresponding eigenvectors of the $n \times n$ matrix whose $j$th column has each entry as $j$.

4. Find all eigenvalues and corresponding eigenvectors of the $n \times n$ matrix whose $j$th row has each entry as $j$.

5. Let $A$ be a $4 \times 4$ non-invertible matrix with all diagonal entries as 1. If $2 + 3i$ is an eigenvalue of $A$, then what are the other eigenvalues?

6. Let $A$ be an $n \times n$ matrix with each eigenvalue having absolute value less than 1. Show that both $I - A$ and $I + A$ are invertible.

7. What is the spectrum of a matrix of rank 1?

8. Let $A \in \mathbb{C}^{n \times n}$ be idempotent, i.e., $A^2 = A$. Show that the only possible eigenvalues of $A$ are 0 and 1.

9. Let $A \in \mathbb{C}^{n \times n}$ be a nilpotent matrix, i.e., $A^k = 0$ for some $k \in \mathbb{N}$. Show that 0 is the only eigenvalue of $A$.

10. Construct a $2 \times 2$ real skew symmetric matrix whose eigenvalues are purely imaginary.

11. Show that if an invertible matrix is real symmetric, then so is its inverse.

12. Suppose $\lambda$ is an eigenvalue of $A$ with an eigenvector $x$, $\mu$ is an eigenvalue of $A^*$ with an eigenvector $y$, and $\overline{\lambda} \neq \mu$. Prove that $x \perp y$.

13. Construct an orthogonal $2 \times 2$ matrix whose determinant is 1.

14. Construct an orthogonal $2 \times 2$ matrix whose determinant is $-1$.

15. Find two different Shur triangularizations of $\begin{bmatrix} 1 & 2 \\ -1 & -2 \end{bmatrix}$.

16. Compute $A^{-1}$ and $A^{10}$ where $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$.

17. Let $A \in \mathbb{F}^{n\times n}$. Let $p(t)$ be a monic polynomial of degree $n$ such that $p(A) = 0$. Does it follow that $p(t) = \chi_A(t)$?

18. Diagonalize the matrix $\begin{bmatrix} 7 & -2 & 0 \\ -2 & 6 & -2 \\ 0 & -2 & 5 \end{bmatrix}$.

19. Is the matrix $\begin{bmatrix} 1 & -10 & 0 \\ -1 & 3 & 1 \\ -1 & 0 & 4 \end{bmatrix}$ diagonalizable?

20. Find an orthogonal matrix that diagonalizes $\begin{bmatrix} -4 & -2 & 2 \\ -2 & -1 & 1 \\ 2 & 1 & -1 \end{bmatrix}$.

21. Let $A$ be a $7\times 7$ matrix with $\chi_A(t) = (t-2)^4(t-3)^3$. It is known that the largest blocks for both the eigenvalues in the Jordan form of $A$ are of order 2. Determine possible Jordan forms of $A$.

22. What is the Jordan form of the $n \times n$ matrix $A$ each row of which is $[1, 2, 3, \ldots, n]$?

23. Show that the positive singular values of $A$ are also singular values of $A^*$. Are there more positive singular values of $A^*$ than those of $A$?

24. Compute the SVD of the matrix $\begin{bmatrix} 1 & 2 & 2 \\ 2 & 0 & 5 \\ 3 & 0 & 0 \end{bmatrix}$.

25. Show that the matrices $\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ and $\begin{bmatrix} 2 & -1 \\ 1 & 0 \end{bmatrix}$ are similar, but they have different singular values.

26. Let $\lambda_1, \ldots, \lambda_n$ be the eigenvalues of an $n \times n$ hermitian matrix $A$. Prove that the singular values of $A$ are $|\lambda_1|, \ldots, |\lambda_n|$.

27. Give an example of a non-hermitian matrix which has an eigenvalue $\lambda$ but $|\lambda|$ is not a singular value of the matrix.

28. Let $A \in \mathbb{C}^{n\times n}$ have eigenvalues $\lambda_1, \ldots, \lambda_n$ and singular values $s_1, \ldots, s_n$. Prove that $|\lambda_1 \cdots \lambda_n| = s_1 \cdots s_n$.

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